



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Homestead at Hog Mountain; Beltan Properties
Location: Dacula, GA
Description: Six Sisters - CBsmt, E-Facing

Date: January 8, 2024
By: JTH / TJH
House Faces: East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	10.0 ft
Floor Area	4,187 ft ²
Volume	40,754 ft ³
Percentage Glass	12 %
Number of Bedrooms	6

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	16038
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.048	Basement	Concrete + 2x6	R-19 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.167	Garage Door		R-6	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.077	Framed		R-13 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.30	Blinds/Draperies	SHGC 0.18
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		44230 Btu/hr		54483 Btu/hr	
Minimum Nominal Capacity		4.91 Tons		15.96 kW	
Load Density		852 SF/Ton		3.8 W/SF	
Floor Area Served		4187.0 SF		4187.0 SF	
Mechanical Ventilation		N/A CFM		N/A CFM	
System Size		5.0 Tons		20.0 kW	
Leaving Air Temperature		57.2 deg F		101.6 deg F	
Installed Capacity		837 SF/Ton		4.8 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Unit 1				
Terrace Level (Basement) Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs & Hall	30	56	43	
Bed 5	58	94	76	
Bed 5 Bath	16	34	25	
Bed 5 WIC	4	6	5	
Closet	3	4	3	
Rec Room	89	65	77	
Dining	10	16	13	
Laundry	37	5	21	
Pantry	4	9	7	
Kitchen	132	36	84	
Bed 4 WIC	10	22	16	
Bed 4 Bath	6	9	7	
Bed 4	55	92	74	
Unfinished	59	134	96	

Unit 1				
Size	5.00	Tons		
Airflow	2000	CFM		
Heat kW	20	kW		
First Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Entry & Living Rm	184	183	183	
Master Bath	50	61	56	
Master WIC (Interior)	8	6	7	
Master Bed	99	91	95	
Master WIC (Rear)	11	20	16	
Breakfast	77	134	106	
Kitchen	177	82	129	
Powder	6	5	5	
Laundry	38	3	21	
Dining	59	103	81	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs (Central)	59	67	63	
Hall	29	42	36	
Bed 2	78	93	86	
Bed 2 Bath	40	42	41	
Bed 2 WIC	12	12	12	
Bed 3 WIC	9	8	9	
Bed 3 Bath	31	43	37	
Bed 3	71	93	82	
-	-	-	-	
Stairs (Garage)	66	89	78	
Kitchen	150	44	97	
Living	162	107	134	
Bath	25	39	32	
Bed	48	50	49	

Cooling Size	1.5	DX Cooling kWh	1507
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Heating Size	1.5	RC Heating kWh	2951
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Best Case	1.5	Total kWh	4458
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SEER 14

Cooling Load Calculation							Ceiling Height		9		9		9		9		9		9		9		9		9		9		9		9		9							
Components					CLTD (°F)	U-Value (Btu/hr-sq-ft-F)	Cooling Load Factor	Entire House		Stairs & Hall		Bed 5		Bed 5 Bath		Bed 5 WIC		Closet		Rec Room		Dining		Laundry		Pantry		Kitchen		Bed 4 WIC		Bed 4 Bath		Bed 4		Unfinished				
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr				
Glass	G1	North	Right	(E)	5 Btu/hr-sf	0	0																																	
	G1	NorthEast		(SE)	7 Btu/hr-sf	0	0																																	
	G1	East	Front	(S)	11 Btu/hr-sf	0	0																																	
	G1	SouthEast		(SW)	12 Btu/hr-sf	0	0																																	
	G1	South	Left	(W)	13 Btu/hr-sf	0	0																																	
	G1	SouthWest		(NW)	22 Btu/hr-sf	0	0																																	
	G1	West	Back	(N)	24 Btu/hr-sf	0	0																																	
	G1	NorthWest		(NE)	15 Btu/hr-sf	0	0																																	
	G1	Sky Light			38 Btu/hr-sf	0	0																																	
	G1	Shaded			5 Btu/hr-sf	74	387																																	
Door	D1	Door	21	0.17	3.51 Btu/hr-sf	5	18	15	79	29.4	154																													
	D2	Door	11	0.17	1.84 Btu/hr-sf	0	0																																	
Above Grade Walls	W1	North	13	0.08	1.00 Btu/hr-sf	326	326	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0																																	
	W1	East	23	0.08	1.77 Btu/hr-sf	311	551	74	131	107	190																													
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0																																	
	W1	South	16	0.08	1.23 Btu/hr-sf	327	402			120	148	60	74																											
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0																																	
	W1	West	23	0.08	1.77 Btu/hr-sf	75	133					70	124																											
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0																																	
	W1	North	13	0.08	1.00 Btu/hr-sf	0	0																																	
Basement Walls	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0																																	
	W1	East	23	0.08	1.77 Btu/hr-sf	0	0																																	
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0																																	
	W1	South	16	0.08	1.23 Btu/hr-sf	0	0																																	
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0																																	
	W1	West	23	0.08	1.77 Btu/hr-sf	0	0																																	
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0																																	
	W2	Grade	0	0.05	0.00 Btu/hr-sf	609	0																																	
	W1	Partition	10	0.08	0.77 Btu/hr-sf	0	0																																	
Other	W1	Garage	30	0.08	2.31 Btu/hr-sf	0	0																																	
	W3	Attic	55	0.05	2.92 Btu/hr-sf	0	0																																	
	W3	Kneewall	55	0.05	2.92 Btu/hr-sf	0	0																																	
	R1	Ceiling	55	0.03	1.43 Btu/hr-sf	0	0																																	
	R2	Vaulted	55	0.05	2.92 Btu/hr-sf	0	0																																	
	F1	Crawl/Uncon.	11	0.08	0.85 Btu/hr-sf	0	0																																	
	F1	Garage	30	0.08	2.31 Btu/hr-sf	0	0																																	
	F2	Slab	0	0.10	0.00 Btu/hr-sf	1522	0	148	-	164	-	52	-	29	-	20	-	236	-	76	-	24	-	29	-	104	-	35	-	44	-	160	-	401	-					
	People				255 Btu/hr	3	765			1	255					1	255															1	255							
Internal	Lights (W)			3.413 Btu/hr-W	763	2604	74	253	82	280	26	89	15	51	10	34	118	403	38	130	12	41	15	51	52	177	18	61	22	75	80	273	201	686						
	Appliances (W)			3.413 Btu/hr-W	1130	3857											250	853			200	683			680	2,321														
	Infiltration	0.25	ACH	0.09 Btu/hr-cf		1233				133										16																				
	Volume			CF	13698																																			
	Area			SF	1522																																			
Subtotal (Btu/hr)							10276																																	
Safety Factor							10 %																																	
Sensible Cooling Load (Btu/hr)							11303																																	
Nominal Tons Required (75% sens. cap.)							1.26																																	
System Selection							1.5 Tons		Space CFM		600		35		68		19		4		3		105		11		43		5		155		12		6		65		69	

HEATING LOAD CALCULATION																																					
Components				Entire House		Stairs & Hall		Bed 5		Bed 5 Bath		Bed 5 WIC		Closet		Rec Room		Dining		Laundry		Pantry		Kitchen		Bed 4 WIC		Bed 4 Bath		Bed 4		Unfinished					
	Type	Description	ΔT	U	A	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr	Q	Btu/hr				
Glass	G1		54	0.30		16.2	Btu/hr-sf	74	1195	15	243	29.4	476	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	29.4	476	0	-		
	G2		54	0.00		0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	G3		54	0.00		0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Door	D1		54	0.17		9.0	Btu/hr-sf	5	45	5	45	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	D2		54	0.17		9.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Walls	W1		54	0.08		4.2	Btu/hr-sf	1039	4319	74.2	309	227	945	130	541	0	-	0	-	72.5	301	0	-	0	-	14.2	59	64.2	267	85	353	0	-	223	928	148.3	617
	W2		54	0.05		2.6	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	W3		54	0.05		2.9	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	W1	Partition	27	0.08		2.1	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	W2	Grade	27	0.05		1.3	Btu/hr-sf	609	789	0	-	0	-	0	-	0	-	72.5	94	0	-	0	-	14.2	18	64.2	83	0	-	0	-	0	-	0	-	458.3	594
Floor/Roof	R1		54	0.03		1.4	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	R2		54	0.05		2.9	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	F1		54	0.08		4.2	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	F2	Slab	27	0.10		2.7	Btu/hr-sf	1522	4109	148	400	164	443	52	140	29	78	20	54	236	637	76	205	24	65	29	78	104	281	35	95	44	119	160	432	401	1,083
Area				SF		1522		148		164		52		29		20		236		76		24		29		104		35		44		160		401			
Infiltration			0.25	ACH	0.24		Btu/hr	0	3329	324		359	114		63	44		516	166		52	63		227	77		96	350		877							
Subtotal (Btu/hr)							13786	1320		2223		795	142		98	1549		371		117	219		858	524		215	2185		3170								
Safety Factor			15 %				2068																														
Heat Loss (Btu/hr)							15854																														
Minimum Heater Size (kW)							4.65																														
System Selection			8	kW	Htg. CFM		600	57		97		35		6	4	67		16		5	10		37	23		9	95		138								

DX Cooling kWh	2091
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By: JTH / TJH

RC Heating kWh	3490
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Level(s): First Floor

Total kWh	5581
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SEER 14

[illegible]

Project:	Homestead at Hog Mountain; Beltan Properties	Date:	January 8, 2024	Typical Ceiling Height	9	Cooling Size	2	DX Cooling kWh	2299
Location:	Dacula, GA	By:	JTH / TJH			Heating Size	2	RC Heating kWh	3700
Description:	Six Sisters - CBsmt, E-Facing	Level(s):	Second Floor	Kneewall Height	1	Best Case	2	Total kWh	6000
	Square Feet	1233	Volume	12736	SF/Ton	617	SEER	14	

COOLING LOAD CALCULATION						Ceiling Height		20		9		9		9		9		9		9		20		9		9		9		9							
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Cooling Load Factor	Entire House		Stairs (Central)		Hall		Bed 2		Bed 2 Bath		Bed 2 WIC		Bed 3 WIC		Bed 3 Bath		Bed 3		Stairs (Garage)		Kitchen		Living		Bath		Bed			
Type	Description							Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr		
Glass	G1 North	Right	(E)	5	Btu/hr-sf	62	327																														
	G1 NorthEast		(SE)	7	Btu/hr-sf	0	0																														
	G1 East	Front	(S)	11	Btu/hr-sf	12	135																														
	G1 SouthEast		(SW)	12	Btu/hr-sf	0	0																														
	G1 South	Left	(W)	13	Btu/hr-sf	38	501																														
	G1 SouthWest		(NW)	22	Btu/hr-sf	0	0																														
	G1 West	Back	(N)	24	Btu/hr-sf	62	1507																														
	G1 NorthWest		(NE)	15	Btu/hr-sf	0	0																														
	G1 Sky Light			38	Btu/hr-sf	0	0																														
D1 Shaded			5	Btu/hr-sf	88	462																															
Door	D1 Door		21	0.17	3.51	Btu/hr-sf	11	39																													
	D2 Door		55	0.17	9.19	Btu/hr-sf	0	0																													
Above Grade Walls	W1 North		13	0.08	1.00	Btu/hr-sf	395	396																													
	W1 NorthEast		19	0.08	1.46	Btu/hr-sf	0	0																													
	W1 East		23	0.08	1.77	Btu/hr-sf	369	654																													
	W1 SouthEast		21	0.08	1.62	Btu/hr-sf	0	0																													
	W1 South		16	0.08	1.23	Btu/hr-sf	382	470																													
	W1 SouthWest		21	0.08	1.62	Btu/hr-sf	0	0																													
	W1 West		23	0.08	1.77	Btu/hr-sf	433	767																													
	W1 NorthWest		19	0.08	1.46	Btu/hr-sf	0	0																													
	W1 North		13	0.08	1.00	Btu/hr-sf	0	0																													
Basement Walls	W1 NorthEast		19	0.08	1.46	Btu/hr-sf	0	0																													
	W1 East		23	0.08	1.77	Btu/hr-sf	0	0																													
	W1 SouthEast		21	0.08	1.62	Btu/hr-sf	0	0																													
	W1 South		16	0.08	1.23	Btu/hr-sf	0	0																													
	W1 SouthWest		21	0.08	1.62	Btu/hr-sf	0	0																													
	W1 West		23	0.08	1.77	Btu/hr-sf	0	0																													
	W1 NorthWest		19	0.08	1.46	Btu/hr-sf	0	0																													
	W2 Grade		0	0.05	0.00	Btu/hr-sf	0	0																													
	W1 Partition		10	0.08	0.77	Btu/hr-sf	0	0																													
Other	W1 Garage		30	0.08	2.31	Btu/hr-sf	126	291																													
	W3 Attic		55	0.05	2.92	Btu/hr-sf	289	841																													
	W3 Kneewall		55	0.05	2.92	Btu/hr-sf	0	0																													
Floor	R1 Ceiling		55	0.03	1.43	Btu/hr-sf	1226	1753																													
	R2 Vaulted		55	0.05	2.92	Btu/hr-sf	0	0																													
	F1 Crawl/Uncon.		11	0.08	0.85	Btu/hr-sf	0	0																													
Internal	F1 Garage		30	0.08	2.31	Btu/hr-sf	34	79																													
	F2 Slab		0	0.10	0.00	Btu/hr-sf	115	0																													
	People				255	Btu/hr	4	1020																													
Vent	Lights (W)				3.413	Btu/hr-W	620	2116																													
	Appliances (W)				3.413	Btu/hr-W	930	3174																													
	Infiltration		0.25	ACH	0.09	Btu/hr-cf		1146																													
Subtotal	Volume				CF		12736																														
	Area				SF		1233																														
	Subtotal (Btu/hr)							15677																													
Safety Factor								10 %																													
Sensible Cooling Load (Btu/hr)																																					
Nominal Tons Required (75% sens. cap.)																																					
System Selection 2 Tons Space CFM																																					
HEATING LOAD CALCULATION																																					
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Heating Load Factor	Entire House		Stairs		Hall		Bed 2		Bed 2 Bath		Bed 2 WIC		Bed 3 WIC		Bed 3 Bath		Bed 3		0		Stairs		Kitchen		Living		Bath		Bed	
Type	Description							Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr		
Glass	G1			54	0.30	16.2	Btu/hr-sf	263	4264	15	243	24	389	45.8	743	21.3	345	0	0	0	0	21.3	345	45.8	743	0	0	25.8	418	0	0	42.7	692	0	0	21.36	346
	G2			54	0.00	0.0	Btu/hr-sf	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
	G3			54	0.00	0.0	Btu/hr-sf	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Door	D1			54	0.17	9.0	Btu/hr-sf	11	100	0	0	6.67	60	0	0	0	0	0	0	0	0	0	0	0	0	0	4.45	40	0	0	0	0	0	0	0		
	D2			54	0.17	9.0	Btu/hr-sf	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
	W1			54	0.08	4.2	Btu/hr-sf	1579	6566	78.7	327	58.5	243	211	877	81.2	338	25.9	107	11.3	47	120	501	209	870	0	0	130	539	140	582	255	1,059	165	686	93.64	389
Walls	W2			54	0.05	2.6	Btu/hr-sf	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	W3			54	0.05	2.9	Btu/hr-sf	289	826	93.7	268	0	0	0	0	42.5	122	25.9	74	11.3	32	0	0	0	0	0	79.5	228	35.9	103	0	0	0	0	0		
	W1 Partition			27	0.08	2.1	Btu/hr-sf	126	262	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	126	262	0	0	0	0	0	0	0		
Floor	R1			27	0.05	1.3	Btu/hr-sf	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	W2			54																																	